1. Define the following terms:*

****(a) Data items:** The smallest named unit of data with meaning, like a single value (e.g., `25`, `'John'`).**

****(b) Meta data:** Data about data; it describes the structure and constraints of the data in the database (e.g., table names, data types).**

****(c) Data:** Raw, unorganized facts and figures collected from the real world (e.g., `25`, `'Ahmedabad'`).**

****(d) Information:** Data that has been processed, organized, and interpreted to be meaningful and useful for decision-making.**

****(e) Entity:** A real-world object or concept (e.g., `Student`, `Course`) about which data can be stored.**

(f) Relationship: A logical association between two or more entities (e.g., a `Student` *enrolls* in a `Course`).

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2. What is Data Abstraction? Explain Three Levels of Data Abstraction.

Data Abstraction hides complex storage details from users. The three levels are:

1. **Physical Level:** Describes *how* data is stored (files, indexes). Hidden from users.

2. **Logical Level:** Describes *what* data is stored and their relationships (tables, attributes). Designers work here.

3. **View Level:** Describes user-specific views of the database, hiding unwanted data.

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3. What is Schema? Explain different types of Schema with example.

A **Schema** is the logical blueprint or structure of the entire database.

* **Physical Schema:** Describes the physical storage (e.g., data stored using a B+ tree index).

* **Logical Schema:** Describes the logical structure (e.g., tables like `Student (id, name)`).

* **View Schema:** Describes user views (e.g., a view showing only student names and grades).

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4. Compare File System vs Database System.

| Aspect | File System | Database System |

| :--- | :--- | :--- |

| **Data Duplication** | High (same data saved many times) | Low (data is shared, less repetition) |

| **Data Accuracy** | Low (can have mismatched info) | High (data is consistent and reliable) |

| **Data Sharing** | Hard (data stuck in separate files) | Easy (all data in one place, easy to access) |

| **Data Rules** | Hard to apply rules | Easy to apply rules (like unique IDs) |

| **Security** | Weak (limited user controls) | Strong (user logins and permissions) |

| **Backup** | Manual (each file separately) | Automatic (full database backup at once) |

| **Ease of Use** | Difficult (need to know file details) | Simple (users just ask for what they need) |

5. Explain Roles and Responsibilities of DBA.

The **Database Administrator (DBA)** manages the DBMS. Key duties include:

* Designing the database schema and granting user permissions.

* Ensuring data security, performing backups, and managing recovery.

* Monitoring system performance and tuning it for efficiency.

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6. What is an Attribute? Explain different types of attributes with example.

An **Attribute** is a property of an entity.

* **Simple:** Atomic value (e.g., `Age`).

* **Composite:** Can be divided (e.g., `Name` into `First_Name`, `Last_Name`).

* **Derived:** Calculated from another attribute (e.g., `Age` from `DOB`).

* **Multi-valued:** Holds multiple values (e.g., `Phone_Numbers`).

* **Key:** Uniquely identifies an entity (e.g., `Roll_Number`).

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7. Explain Specialization and Generalization with example.

* **Specialization:** A top-down process of dividing a general entity into specialized sub-entities (e.g., `Employee` → `Teacher`, `Clerk`).

* **Generalization:** A bottom-up process of combining common features of specific entities into a general entity (e.g., `Car`, `Bike` → `Vehicle`).

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8. Demonstrate Mapping Cardinality with diagram.

Mapping Cardinality defines relationship participation.

* **One-to-One (1:1):** One `Person` has one `Passport`.

  `[Person]1----1[Passport]`

* **One-to-Many (1:M):** One `Department` has many `Employees`.

  `[Department]1----M[Employee]`

* **Many-to-One (M:1):** Many `Employees` work in one `Department`.

  `[Employee]M----1[Department]`

* **Many-to-Many (M:N):** Many `Students` take many `Courses`.

  `[Student]M----N[Course]` --

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10. Define the SQL data types.

SQL data types define the kind of data a column can hold. Common types are:

* **Numeric:** `INT` (integer), `FLOAT` (floating-point number), `NUMBER(n,d)` (decimal number).

* **Character/String:** `CHAR(n)` (fixed-length string), `VARCHAR(n)` (variable-length string).

* **Date/Time:** `DATE` (stores date: YYYY-MM-DD), `TIME` (stores time: HH:MM:SS), `TIMESTAMP` (stores both date and time).

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11. Explain the DDL command in SQL.

DDL (Data Definition Language) commands define the database structure.

* **CREATE** (build objects), **ALTER** (modify structure), **DROP** (delete objects), **TRUNCATE** (remove all data from a table).

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12. Explain the DML command in SQL.

DML (Data Manipulation Language) commands manage data within tables.

* **SELECT** (retrieve data), **INSERT** (add new data), **UPDATE** (modify data), **DELETE** (remove data).

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13. Write SQL queries for the following tables:

T1 (Empno, Ename, Salary, Designation), T2 (Empno, Deptno)

1. **Show all employees:**

   ```sql
   SELECT * FROM T1;
   ```

2. **Join to show Empno, Ename, Deptno:**

   ```sql
   SELECT T1.Empno, T1.Ename, T2.Deptno
   FROM T1 INNER JOIN T2 ON T1.Empno = T2.Empno;
   ```

```

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### **14. Write answers for the given queries:**

**stud_master (enroll_no, name, city, dept)**

1. **Show all records:**

```sql
SELECT * FROM stud_master;
```

2. **Show students from 'Surat':**

```sql
SELECT * FROM stud_master WHERE city = 'Surat';
```

3. **Show names of 'Computer' department students:**

```sql

**SELECT name FROM stud_master WHERE dept = 'Computer';**

```
```